The *syntheticTurbulenceInlet* boundary condition for OpenFoam-12: method and formal description

Hursanay Fyhn and Eirik Holm Fyhn
*SINTEF Energy Research**
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This document presents the mathematical formulation of *syntheticTurbulenceInlet*, a velocity inlet boundary condition for OpenFoam-12 that synthesises turbulent fluctuations from a target Reynolds-stress tensor and integral-length-scale field using a divergence-free synthetic eddy method (DFSEM). The present implementation derives from the ESI *turbulentDFSEMInlet*, with three types of improvement. First, two equation-level corrections bring the boundary condition into line with the original DFSEM paper [1], from which the original DFSEM code had deviated. Second, the global normalisation constant is re-derived from the eddy-shape integrals; the new constant replaces the per-Γ scaling used both in [1] and in the original DFSEM code. Third, the applicability is broadened through geometry-aware sampling, automatic profile mapping, runtime-selectable inlet targets, swirl, periodic and wall-image ghost eddies, and reproducible random number generation, all of which are absent from the inspiration code. The combined effect is improved fidelity and a substantially reduced configuration burden on the user.

CONTENTS

I. INTRODUCTION

For scale-resolving simulations such as Large-Eddy Simulation (LES) or hybrid RANS–LES, the inlet must supply turbulent fluctuations consistent with prescribed turbulence statistics. The Synthetic Eddy Method (SEM) realises this by populating a control volume centred on the inlet (the *eddy box*) with N localised eddies. Each eddy carries a length scale, a sign vector and a Reynolds-stress signature; their superposed contributions on the inlet patch reconstruct an instantaneous velocity field whose first and second moments match the target. The divergence-free variant DFSEM [1] ensures that the synthesised field has zero divergence by construction, which avoids spurious pressure transients downstream.

The present implementation derives from the ESI `turbulentDFSEMInlet`. For each topic below, the mathematical formulation as implemented is given first, followed (where applicable) by a comparison that identifies how the present version differs from the inspiration code and, where relevant, from the original DFSEM paper [1]. Topics that have no counterpart in either are flagged as *new*.

Users employing this boundary condition in academic work are kindly asked to cite the companion papers [2, 3], both submitted to the *Journal of Engineering for Gas Turbines and Power*.

II. METHOD AND IMPROVEMENTS

A. Velocity decomposition

The instantaneous patch velocity is decomposed as

$$\boldsymbol{U}(\boldsymbol{x},t) = \boldsymbol{U}_{\text{mean}}(\boldsymbol{x},t) + \boldsymbol{U}'(\boldsymbol{x},t), \tag{1}$$

where $\boldsymbol{U}_{\text{mean}}$ follows a wall-normal mean profile (Section II G) and $\boldsymbol{U}'$ is the synthetic fluctuation generated by the eddy ensemble (Section II C). After construction, the patch field is rescaled multiplicatively to enforce the user-prescribed mass or volumetric flow rate (Section II L).

* hursanay.fyhn@sintef.no

B. Eddy parameters

An eddy k is characterised by:

- its centre $\boldsymbol{x}_k = \boldsymbol{x}_{0,k} + x_k \boldsymbol{n}$, where $\boldsymbol{x}_{0,k}$ is an in-plane reference position drawn uniformly over the patch geometry, $\boldsymbol{n}$ is the patch normal pointing into the domain, and $x_k \in (-\sigma_{\max}, \sigma_{\max})$ is the streamwise offset within the eddy box;
- integral-length scales $\boldsymbol{\sigma}_k = (\sigma_1, \sigma_2, \sigma_3)_k$ defined in its principal frame, with σ_3 the major axis (aligned with the eigenvector of R of largest eigenvalue) set by the local integral length scale (Section II G), and $\sigma_1 = \sigma_2 = \sigma_3/\gamma$ (Section II D); the patch-wide maximum $\sigma_{\max} = \max_k \sigma_{3,k}$ sets the eddy-box half-depth;
- a sign vector $\boldsymbol{\epsilon}_k \in \{-1, +1\}^3$;
- a rotation matrix Q_k from principal to global frame whose columns are the eigenvectors of the local Reynolds-stress tensor R.

Eddies are seeded inside the box until the eddy-density target $\sum_k V_k/V_0 \geq d$ is reached, with $V_k = \frac{4}{3}\pi\sigma_1\sigma_2\sigma_3$, eddy-box volume $V_0 = 2A\sigma_{\max}$, A the inlet area, and d a user parameter (default $d = 1$).

C. Eddy shape and fluctuation field

For an evaluation point $\boldsymbol{x}$ on the patch and an eddy k, define the relative position in the eddy principal frame

$$\boldsymbol{r}_p = \mathrm{cmptDivide}\big(\mathsf{Q}_k^{\mathsf{T}}(\boldsymbol{x} - \boldsymbol{x}_k), \boldsymbol{\sigma}_k\big)\,. \tag{2}$$

Eddies with $|\boldsymbol{r}_p| \geq 1$ contribute nothing. Otherwise, the fluctuation contribution is

$$\boldsymbol{U}'_k(\boldsymbol{x}) = c_{1,k}\, \mathsf{Q}_k \,(1 - |\boldsymbol{r}_p|^2)\, \big[\boldsymbol{\sigma}_k \odot (\boldsymbol{r}_p \times \boldsymbol{\alpha}_k)\big], \tag{3}$$

where $\odot$ is the Hadamard (component-wise) product and the cross product is the standard one. Per component in the eddy principal frame this reads $\tilde{U}'_{k,\beta} = c_{1,k}\,(1 - |\boldsymbol{r}_p|^2)\sigma_{k,\beta}(\boldsymbol{r}_p \times \boldsymbol{\alpha}_k)_\beta$, with the global-frame value $\boldsymbol{U}'_k = \mathsf{Q}_k\,\tilde{\boldsymbol{U}}'_k$. The intensity $\boldsymbol{\alpha}_k$ is determined by the local Reynolds-stress signature (Section II D), and the per-eddy and global normalisation constants $c_{1,k}$ and C_2 are specified in Section II E; their values are derived in Appendix A. The total fluctuation is

$$\boldsymbol{U}'(\boldsymbol{x}) = C_2 \sum_{k=1}^{N} \boldsymbol{U}'_k(\boldsymbol{x}). \tag{4}$$

Improvement over the original DFSEM code. Two equation-level issues in the original DFSEM code are addressed in Equations (2) and (3).

Order of normalisation and rotation. The original DFSEM code normalises the relative position component-wise in the *global* frame and only afterwards rotates it into the principal frame: $\boldsymbol{r} = \mathrm{cmptDivide}(\boldsymbol{x} - \boldsymbol{x}_k, \boldsymbol{\sigma}_k)$ and then $\boldsymbol{r}_p = \mathsf{Q}_k^{\mathsf{T}}\boldsymbol{r}$. Because $\boldsymbol{\sigma}_k$ is defined per component in the *principal* frame, this mixes axes whenever the principal frame is not aligned with the global frame, so the eddy is not in fact elongated along its designated major axis. Equation (2) rotates first and then normalises, restoring the correct elongation along the eigenvector of R of largest eigenvalue (i.e. along the streamwise direction in typical wall-bounded flows). The corrected order matches the formulation of [1].

Scalar versus per-component shape. The original DFSEM code evaluates the shape function component-wise as $q_i = \sigma_i(1 - r_{p,i}^2)$, with $r_{p,i}$ the principal-frame *components* of the relative position. Since the eddy has unit spherical support $|\boldsymbol{r}_p| < 1$ (used in the angular integrals of Appendix A), the shape factor must depend on the scalar radius $|\boldsymbol{r}_p|$ only: $\sigma_i(1 - |\boldsymbol{r}_p|^2)$. The scalar form, used in Equation (3), also matches [1].

D. Reynolds-stress mapping

The local Reynolds-stress tensor R at the eddy seed position is diagonalised:

$$\mathsf{R}\,\boldsymbol{e}_i = \lambda_i\,\boldsymbol{e}_i, \qquad \lambda_1 \leq \lambda_2 \leq \lambda_3, \tag{5}$$

and the principal-to-global rotation $\mathsf{Q} = [\boldsymbol{e}_1\ \boldsymbol{e}_2\ \boldsymbol{e}_3]$ satisfies $\mathsf{Q}^{\mathsf{T}}\,\mathsf{R}\,\mathsf{Q} = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)$. The eddy major axis is aligned with $\boldsymbol{e}_3$; the remaining two scales are linked by an isotropy ratio γ chosen from $\{1, \sqrt{2}, \ldots, \sqrt{8}\}$:

$$\sigma_1 = \sigma_2 = \sigma_3/\gamma. \tag{6}$$

The principal-frame intensity components are obtained from a diagonalised second-moment constraint:

$$\alpha_\beta^2 = -\frac{\lambda_\beta}{\sigma_\beta^2} + \frac{\lambda_{\beta+1}}{\sigma_{\beta+1}^2} + \frac{\lambda_{\beta+2}}{\sigma_{\beta+2}^2}, \qquad \beta = 0, 1, 2 \pmod 3, \tag{7}$$

with sign assignment $\alpha_\beta \leftarrow \epsilon_\beta\sqrt{\alpha_\beta^2}$. If any $\alpha_\beta^2 < 0$ for the trial γ, the next value is tried. After a successful assignment, the trial list is reordered by the fixed 8-cycle permutation $[2, 8, 6, 7, 4, 5, 1, 3]$ so that successive eddies start their search from different γ values; this deterministically distributes the isotropy ratios across the ensemble without an extra random draw.

The local R itself is sampled from a wall-normal lookup table at the seed point's wall distance and rotated from the wall-aligned frame $(\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w})$ to the global frame:

$$\mathsf{R}_{\mathrm{global}} = \mathsf{T}^{\mathsf{T}}\mathsf{R}_{\mathrm{wall}}\mathsf{T}, \qquad \mathsf{T} = [\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}], \tag{8}$$

where $\boldsymbol{u} = \boldsymbol{n}$, $\boldsymbol{v}$ is the unit vector pointing from the seed toward the nearest wall, and $\boldsymbol{w} = \boldsymbol{u} \times \boldsymbol{v}$.

E. Normalisation constants

Two normalisation factors enter Equations (3) and (4): a per-eddy factor $c_{1,k}$ that depends on the eddy's own dimensions, and a global factor C_2 that depends on the eddy box and the eddy count. In the present implementation,

$$c_{1,k} = \frac{1}{\sqrt{V_k}}, \qquad V_k = \tfrac{4}{3}\pi\, \sigma_{1,k}\sigma_{2,k}\sigma_{3,k}, \tag{9}$$

and

$$\boxed{C_2 \;=\; s\,\sqrt{\frac{315}{16}\,\frac{V_0}{N}}} \tag{10}$$

where s is a user-tuneable scaling prefactor (default $s = 1$) retained for empirical adjustment.

Improvement over the original DFSEM code and the paper. The original DFSEM code combines a per-eddy factor $c_1 = \bar{\sigma}\,\sigma_{\min}/(\sigma_x\sigma_y\sigma_z)$ with a global factor $c = (10\,V_0)^m/\sqrt{N}$ at $m = 1/2$, and applies the per-Γ table $C_2(\Gamma) \in \{2, 1.875, 1.737, 1.75, 0.91, 0.825, 0.806, 1.5\}$ as a $1/(2C_2)$ scaling on the squared principal-frame intensity. Together these reproduce the normalisation reported in [1] (Eq. 11 and Table 1 there) exactly. We re-derived the variance-matching constraint from the eddy-shape integrals and obtained different scaling factors: the per-eddy constant becomes the geometric weight (9), the per-Γ table is removed and is absorbed into the global factor, and the global constant becomes Equation (10). The rederived prefactor on $\sqrt{V_0/N}$ is $\sqrt{315/16} \approx 4.44$, against the original DFSEM code value $\sqrt{10} \approx 3.16$.

F. Eddy convection and respawn

Each time step the eddies are translated downstream by $\Delta t\,\bar{U}$. Eddies whose streamwise offset exceeds $\sigma_{\max}$ are respawned with new random in-plane positions and intensities, and a streamwise offset

$$x_k^{\mathrm{new}} = x_k - 2\lfloor x_k/\sigma_{\max}\rfloor\,\sigma_{\max}. \tag{11}$$

Improvement over the original DFSEM code. The original DFSEM code respawns at $x^{\mathrm{new}} = x - \lfloor x/\sigma_{\max}\rfloor\sigma_{\max} \in [0, \sigma_{\max})$. Although the initial seeding samples $(-\sigma_{\max}, \sigma_{\max})$, the steady-state distribution collapses after a transient to the upstream half of the box only. Equation (11) respawns symmetrically across the full box, preserving the original distribution and removing the transient bias.

G. Mean velocity, Reynolds stress, and length scale from DNS profiles

The mean velocity on the patch follows the boundary-layer profile

$$\boldsymbol{U}_{\mathrm{mean}}(\boldsymbol{x}) = \boldsymbol{n}\,\frac{U^+\big(y^+(\boldsymbol{x})\big)}{U_{\mathrm{avg}}}\,\bar{U}, \tag{12}$$

where $U^+(y^+)$ is interpolated from a tabulated DNS profile parameterised by the wall-distance Reynolds number

$$y^+ = y\,U_\tau/\nu, \tag{13}$$

y is the local wall distance (Section III), $\bar{U}$ is the area-averaged target speed, and $U_{\mathrm{avg}} = \int U^+\,\mathrm{d}A/A$ normalises the profile so that the flux constraint yields $\bar{U}$ before fluctuations are added. The Reynolds-stress tensor R and integral-length scale L are sampled from analogous tables at the same y^+, with the table value L^+ converted to a physical length by $L = L^+\,\nu/(U_\tau\,\mathtt{Lref})$, where `Lref` is a user-tunable rescaling (default 1).

The major-axis length σ_3 assigned to each seeded eddy follows from this local L by clipping against the mixing-length cap and a grid-resolution floor:

$$\sigma_3 \;=\; \mathrm{clip}\big(|L|;\ \mathtt{nCellPerEddy}\cdot\Delta_{\max},\ \kappa\delta\big), \tag{14}$$

with $\kappa = 0.41$, δ the half wall-to-wall distance, and $\Delta_{\max}$ is the longest characteristic dimension over all patch-adjacent cells.

New. DNS reference profiles for $U^+(y^+)$, $L^+(y^+)$ and the six components of $\mathsf{R}(y^+)$ are bundled with the source. The original DFSEM code requires the user to supply these as `PatchFunction1` entries. Here the user needs to supply only the inlet target ($\bar{U}$ or $\dot{m}$) and, optionally, U_τ, with spatial profiles recovered automatically.

H. Friction velocity

If the user does not supply U_τ, it is computed from the bulk Reynolds number

$$\mathrm{Re} = \bar{U}\,D/\nu, \tag{15}$$

where D is the wall-to-wall distance for the detected geometry. For $\mathrm{Re} < 3000$ the laminar branch $\sqrt{f} = 8/\sqrt{\mathrm{Re}}$ is used; otherwise a smooth-pipe correlation is solved for $\sqrt{f}$ by Newton iteration:

$$\frac{1}{\sqrt{f}} \;=\; 1.93\,\log_{10}\!\big(\mathrm{Re}\sqrt{f}\big) - 0.537, \tag{16}$$

and $U_\tau = \bar{U}\sqrt{f/8}$. A lower bound $U_{\tau,\min}$, derived from the outermost row of the DNS table so that the entire profile fits in the available wall-to-wall distance, is enforced. A user-supplied U_τ above this bound thins the near-wall layer accordingly: the DNS profile is mapped only over $y^+ \le y^+_{\max}$, and the patch interior beyond that point is filled with the half-width value of the DNS table.

New. The original DFSEM code has no concept of friction velocity. Auto-computation from Re removes one input the user would otherwise have to estimate themselves, and an explicit U_τ becomes a direct knob on the inlet boundary-layer thickness: $U_{\tau,\min}$ stretches the full DNS profile across the entire wall-to-wall distance, while larger values produce a correspondingly thinner near-wall layer with a flat-topped interior.

I. Geometry detection and wall distance

The patch geometry is identified by the user via `geometryMode`, chosen from `annulus`, `disk`, `rectangle`, `channel`, or `arbitrary`. The patch normal, in-plane basis, centre and characteristic dimensions (ℓ_1, ℓ_2) are then extracted automatically from the mesh. For each eddy seed point, the wall-normal distance y used in Equation (13) is computed as the distance to the nearest wall:

- annulus: $\min(r - \ell_1,\, \ell_2 - r)$;
- disk: $\ell_2 - r$;
- rectangle: $\min(\ell_1/2 - |\xi_1|,\, \ell_2/2 - |\xi_2|)$, where $\xi_i = (\boldsymbol{x} - \boldsymbol{x}_c) \cdot \boldsymbol{e}_i$ are the signed in-plane offsets from the patch centre along $\boldsymbol{e}_1$ (the short axis, $\ell_1 \leq \ell_2$) and $\boldsymbol{e}_2$ (the long axis);
- channel: $\ell_i/2 - |\xi_i|$ for the wall-bounded direction ($i = 1$ if `wallLongOrShort=long`, $i = 2$ otherwise).

The first four modes provide wall-aware sampling that is consistent with the DNS lookup tables of Section II G; the `arbitrary` mode generates fluctuations on patches that do not match a canonical shape, using uniform mean velocity and constant R, L from the outermost table row, corresponding to the channel centreline of the reference dataset.

New. The original DFSEM code is geometry-agnostic and requires the user to supply all profiles as `PatchFunction1` fields with no notion of wall distance. Built-in detection of the canonical shapes lets the boundary condition compute wall distance, characteristic length and area sampling from the mesh alone. The mesh-derived patch normal $\boldsymbol{n}$ also fixes the mean-flow direction in Equation (12) and the eddy-box orientation in Equation (2), so the boundary condition operates correctly for inlets at any orientation without the user prescribing a flow direction.

J. Wall-image and periodic ghost eddies

Eddies near a wall would otherwise contribute one-sidedly to the near-wall second moments. To restore symmetric coverage, the eddy-summation routine adds image eddies as follows:

- rectangle, channel: reflections at $\pm\ell_1\boldsymbol{e}_1$ and $\pm\ell_2\boldsymbol{e}_2$ (the latter periodic for channel mode);
- disk: a single reflection at $-2\ell_2\hat{r}$;
- annulus: reflections at $\pm 2\ell_1\hat{r}$ and $\pm 2\ell_2\hat{r}$.

New. Not present in the original DFSEM code.

K. Swirl

For `annulus` and `disk`, an azimuthal velocity is optionally added. The swirl number is defined as

$$S = \frac{\int_{r_i}^{r_o} \rho\, u_x\, u_\theta\, r^2\, \mathrm{d}r}{\bar{r} \int_{r_i}^{r_o} \rho\, u_x^2\, r\, \mathrm{d}r}, \qquad \bar{r} = (r_i + r_o)/2. \tag{17}$$

To keep u_θ smooth at $r_i = 0$ and shaped like u_x, we set

$$u_\theta(r) = f_\theta\, r\, u_x(r), \tag{18}$$

in which case Equations (17) and (18) yield

$$f_\theta = S\, \bar{r}\, \frac{\displaystyle\int_{r_i}^{r_o} \rho\, u_x^2\, r\, \mathrm{d}r}{\displaystyle\int_{r_i}^{r_o} \rho\, u_x^2\, r^3\, \mathrm{d}r}. \tag{19}$$

The integrals are evaluated as face sums weighted by $|\boldsymbol{S}_f|$, the face area vector (magnitude equal to face area, direction along the outward normal).

New. Not present in the original DFSEM code.

L. Flow-rate enforcement

After superposition of the eddy contributions, the patch field is rescaled multiplicatively to enforce either the mean-velocity or the mass-flow target:

$$\boldsymbol{U} \leftarrow \boldsymbol{U} \cdot \Phi_{\text{target}} \cdot f_{\text{Corr}},$$
$$f_{\text{Corr}} = \begin{cases} \dfrac{A}{\displaystyle\sum_f \boldsymbol{U}_f \cdot (-\boldsymbol{S}_f)} & \text{(mean-velocity mode)}, \\[2ex] \dfrac{1}{\displaystyle\sum_f \rho_f\, \boldsymbol{U}_f \cdot (-\boldsymbol{S}_f)} & \text{(mass-flow mode)}. \end{cases} \tag{20}$$

Φ_{target} is supplied via a `Function1<scalar>` (constant, table, polynomial, sine, coded, etc.), so the inlet target may vary in time. The flag `massFlowMode` switches between the $\bar{U}$ and $\dot{m}$ targets. The density ρ_f is the patch density in the compressible case and a constant `rhoInf` in the incompressible case; the branch is chosen automatically from the dimensions of the volumetric/mass flux ϕ.

New. The original DFSEM code rescales to a fixed bulk velocity inferred from the user-supplied mean-velocity field. Time-varying targets, the `massFlowMode` toggle and the compressible/incompressible auto-detection are added here.

M. Reproducible random number generator

The Mersenne Twister generator is seeded from a user-supplied integer (default 536), independently of process rank.

New. The original DFSEM code seeds from the process rank only, so the eddy ensemble depends on the rank assignment: changing the number of MPI processes or the partitioning gives a statistically different ensemble. The user-supplied seed used here is identical on every process, so the eddy ensemble is the same regardless of decomposition.

III. TEST RESULTS

Figure 1 shows time-averaged turbulence statistics measured along streamwise lines at several stations downstream of the inlet for a canonical channel flow. Profiles develop rapidly toward the DNS target The new normalisation constant in Equation (10) is the dominant contribution to the improved match relative to the original DFSEM code.

IV. USAGE

Once the source repository has been obtained:

1. Source your local OpenFOAM-12 environment as usual (e.g. `source etc/bashrc` in your OpenFOAM tree).
2. Compile the boundary condition with `wmake` from the project root.
3. Load the resulting library in your case's `system/controlDict`:

```
libs ("libsyntheticTurbulenceInlet.so");
```

Ready-to-run cases for each `geometryMode` are supplied under `tutorials/`. Table I lists the dictionary entries accepted by the boundary condition.

A minimal patch entry is

```
inlet
{
    type    syntheticTurbulenceInlet;
    geometryMode annulus;
    massFlowMode true;
    massFlow   2.53e-4;
}
```

Appendix A: Derivation of the principal-frame intensity and normalisation constants

This appendix shows that the principal-frame intensity Equation (7), together with the constants $c_{1,k}$ in Equation (9) and C_2 in Equation (10), are the unique choice that makes the eddy ensemble reproduce the target Reynolds tensor R.

The derivation works entirely in the eddy principal frame. The global second moment is the orthogonal similarity

$$\langle u_i' u_j' \rangle = \mathsf{Q}_{ik} \, \langle \tilde{u}_k' \tilde{u}_l' \rangle \, \mathsf{Q}_{lj}^{\mathsf{T}}, \tag{A1}$$

so requiring $\langle \tilde{u}_a' \tilde{u}_b' \rangle = \lambda_a \delta_{ab}$ in the principal frame is equivalent to recovering R in the global frame, with $\lambda_1 \le \lambda_2 \le \lambda_3$ the eigenvalues of Equation (5).

Starting ansatz. Write the principal-frame fluctuation in terms of a single, as-yet-unfixed per-eddy intensity $\tilde{\boldsymbol{\alpha}}^k$ that absorbs all the eventual normalisation:

$$\begin{gathered} \tilde{u}_\beta'(\boldsymbol{x}) = \sum_{k=1}^{N} \sigma_\beta^k \, q(d^k) \, (\boldsymbol{r}_p^{\,k} \times \tilde{\boldsymbol{\alpha}}^k)_\beta, \\ q(d) = \begin{cases} 1 - d^2 & d < 1, \\ 0 & d \ge 1, \end{cases} \end{gathered} \tag{A2}$$

with $d^k = |\boldsymbol{r}_p^{\,k}|$ and $\boldsymbol{r}_p^{\,k}$ from Equation (2). The tilde marks $\tilde{\boldsymbol{\alpha}}$ as the variable used inside this appendix's bookkeeping; once the variance-matching constraint determines it, the geometric prefactor is extracted into $c_{1,k}$ and C_2 and the tilde is dropped to recover the clean $\boldsymbol{\alpha}$ of the main text (Equation (7)). The $\boldsymbol{\sigma}^k$ and $|\tilde{\boldsymbol{\alpha}}^k|$ are fixed for each eddy and only the signs of the three components of $\tilde{\boldsymbol{\alpha}}^k$ are random (independent and uniform over $\{-1, +1\}^3$, both within an eddy and across eddies).

Sign average. Let $\langle \cdot \rangle_\alpha$ denote averaging over the random signs. Component-wise sign independence gives $\langle \tilde{\alpha}_i^k \tilde{\alpha}_j^l \rangle_\alpha = \delta_{kl} \delta_{ij} (\tilde{\alpha}_i^k)^2$. Applied to the x-component of Equation (A2) squared, the cross terms between $\tilde{\alpha}_y$ and $\tilde{\alpha}_z$ vanish and the diagonal in kl collapses the double sum:

$$\begin{aligned} \langle \tilde{u}_x' \tilde{u}_x' \rangle_\alpha = \sum_k (\sigma_x^k)^2 \left[q(d^k) \right]^2 \big[&(r_{p,y}^k)^2 (\tilde{\alpha}_z^k)^2 \\ &+ (r_{p,z}^k)^2 (\tilde{\alpha}_y^k)^2 \big]. \end{aligned} \tag{A3}$$

Position average. Let $\langle \cdot \rangle_r$ denote averaging over uniform eddy positions in the eddy box of volume $V_0 = 2A\sigma_{\max}$. Assume further that $\boldsymbol{\sigma}^k$ and $|\tilde{\boldsymbol{\alpha}}^k|$ are slowly varying on the scale of an eddy, so they may be evaluated at the field point and pulled outside the position integral; this is exact only when eddies are much smaller than the scales over which R and L vary. Then all N terms in Equation (A3) are statistically identical and the sum collapses to N times one position integral:

$$\langle \tilde{u}_x' \tilde{u}_x' \rangle = \frac{N \sigma_x^2}{V_0} \int_{V_0} \mathrm{d}^3 x_k \, [q(d)]^2 \big(r_{p,y}^2 \tilde{\alpha}_z^2 + r_{p,z}^2 \tilde{\alpha}_y^2 \big). \tag{A4}$$

Change of variables and angular integration. Using $r_{p,i} = (x_{k,i} - x_{p,i})/\sigma_i$ componentwise, $\mathrm{d}^3 x_k = \sigma_x \sigma_y \sigma_z \, \mathrm{d}^3 r_p$, and the integrand is supported on $|\boldsymbol{r}_p| < 1$. In spherical coordinates (r, θ, ϕ) with $r_{p,y} = r \sin\theta \sin\phi$ and $r_{p,z} = r\cos\theta$,

$$\begin{aligned} \langle \tilde{u}_x' \tilde{u}_x' \rangle = \frac{N \sigma_x^2 \, \sigma_x \sigma_y \sigma_z}{V_0} &\int_0^1 r^4 (1 - r^2)^2 \mathrm{d}r \\ \times \int_0^\pi \sin\theta \mathrm{d}\theta &\int_0^{2\pi} \mathrm{d}\phi \big(\tilde{\alpha}_z^2 \sin^2\theta \sin^2\phi + \tilde{\alpha}_y^2 \cos^2\theta \big). \end{aligned} \tag{A5}$$

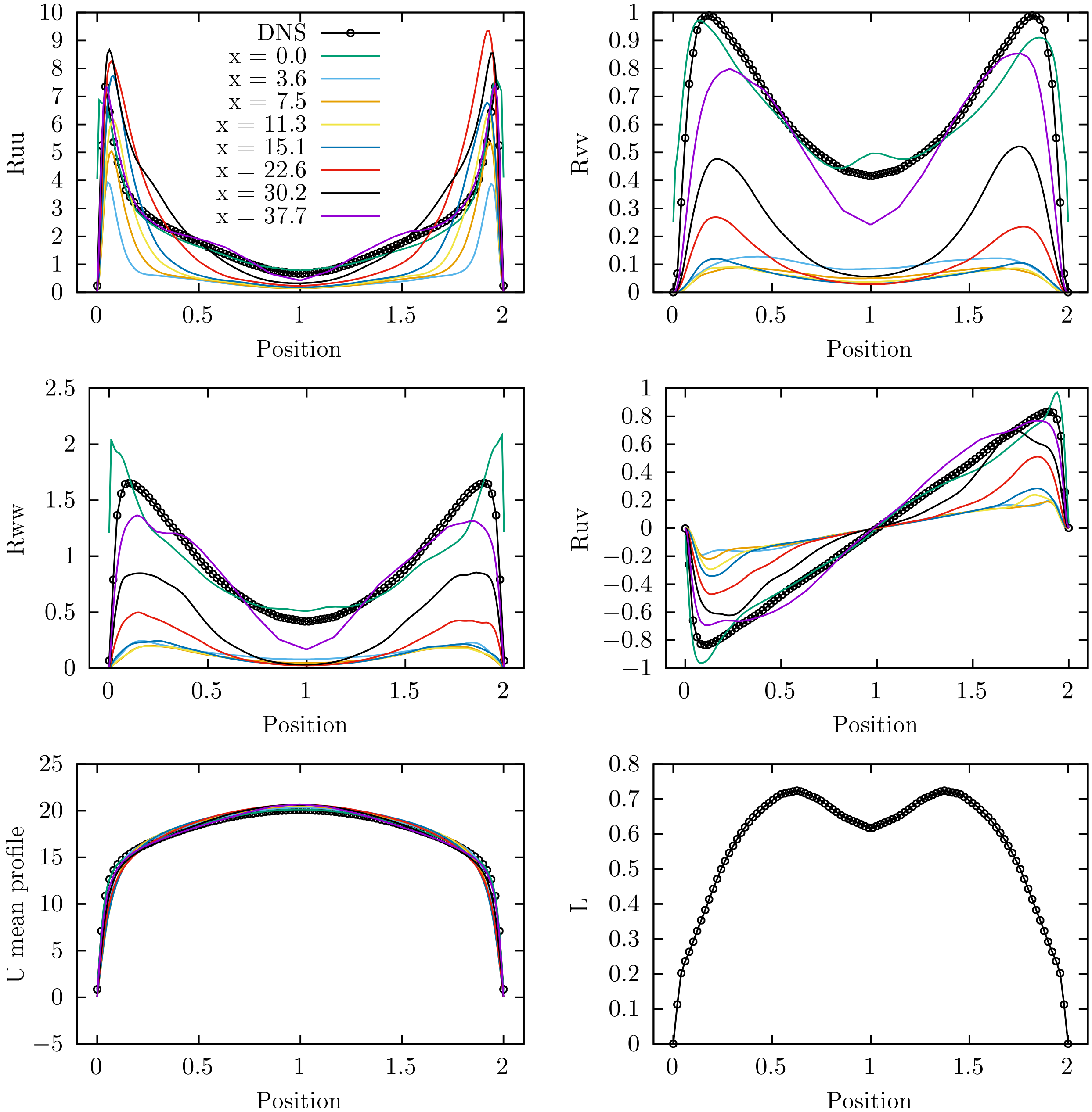

FIG. 1. Time-averaged U, R and L along streamwise lines at various positions x, compared against the DNS target.

With

$$\int_0^1 r^4(1-r^2)^2\,\mathrm{d}r = \tfrac{8}{315}, \tag{A6}$$

$$\int_0^\pi \sin^3\theta\,\mathrm{d}\theta \int_0^{2\pi} \sin^2\phi\,\mathrm{d}\phi = \tfrac{4}{3}\,\pi, \tag{A7}$$

$$\int_0^\pi \sin\theta\cos^2\theta\,\mathrm{d}\theta \int_0^{2\pi} \mathrm{d}\phi = \tfrac{2}{3}\,(2\pi), \tag{A8}$$

both angular integrals reduce to $4\pi/3$ and Equation (A5) becomes

$$\langle \tilde{u}'_x \tilde{u}'_x \rangle = \underbrace{\frac{8}{315}\,\frac{NV_e}{V_0}\,\sigma_x^2}_{\equiv\,\tilde{C}_x}(\tilde{\alpha}_y^2 + \tilde{\alpha}_z^2) \;\stackrel{!}{=}\; \lambda_x, \tag{A9}$$

where $V_e = \frac{4}{3}\pi\sigma_x\sigma_y\sigma_z$ is the eddy ellipsoid volume.

Off-diagonal cross-correlations. For Equation (A9) to suffice it must also be the case that $\langle \tilde{u}'_i \tilde{u}'_j \rangle = 0$ for $i \neq j$ (otherwise the matching condition is incomplete). Consider $\langle \tilde{u}'_x \tilde{u}'_y \rangle$. Expanding $\tilde{u}'_x = \sum_k \sigma_x^k q^k (r_{p,y}^k \tilde{\alpha}_z^k - r_{p,z}^k \tilde{\alpha}_y^k)$

TABLE I. Dictionary entries accepted by `syntheticTurbulenceInlet`.

Key	Type	Default	Description
Mandatory			
`geometryMode`	word	—	One of `annulus`, `disk`, `rectangle`, `channel`, `arbitrary`.
`Umean` *or* `massFlow`	`Function1<scalar>`	—	Inlet target. `Umean` is read if `massFlowMode=false`, `massFlow` otherwise.
Optional			
`massFlowMode`	bool	`false`	`true`: read `massFlow` (kg/s). `false`: read `Umean` (m/s).
`wallLongOrShort`	word	`long`	Channel only: which side of the rectangle carries walls; the other is treated as periodic.
`arbitraryModeRadius`	scalar	max patch radius	Arbitrary mode only: radius within which to seed eddies.
`arbitraryModeCenter`	vector	patch centre	Arbitrary mode only: centre point for eddy seeding.
`Utau`	scalar	auto from Eq. (16)	Friction velocity. A negative value forces the lower limit $U_{\tau,\min}$, which maps the entire DNS profile across the available wall-to-wall distance.
`swirlNr`	scalar	`0`	Swirl number S. Annulus and disk only.
`d`	scalar	`1`	Eddy fill fraction $\sum V_k/V_0$.
`scale`	scalar	`1` (`0.2` for `arbitrary`)	Heuristic prefactor s on C_2 (Equation (10)).
`nCellPerEddy`	label	`1` (`5` for `arbitrary`)	Minimum eddy length in cell counts.
`Lref`	scalar	`1`	Length-scale normalisation for L.
`seed`	uint	`536`	Mersenne-Twister seed.
`rhoInf`	scalar	`1`	Reference density used by the incompressible mass-flow rescaling in Equation (20).

and the analogue with σ_y^k for $\tilde{u}'_y$, the product expands into four bilinears in components of $\tilde{\boldsymbol{\alpha}}$. Three mix *distinct* components ($\tilde{\alpha}_z\tilde{\alpha}_x$, $\tilde{\alpha}_y\tilde{\alpha}_x$, $\tilde{\alpha}_y\tilde{\alpha}_z$) and vanish under the sign average; only the bilinear in $\tilde{\alpha}_z\tilde{\alpha}_z$ survives. After the sign average collapses $kl \to k$ this leaves

$$\langle \tilde{u}'_x \tilde{u}'_y \rangle_\alpha = -\sum_k \sigma_x^k \sigma_y^k [q(d^k)]^2 r_{p,x}^k \, r_{p,y}^k \, (\tilde{\alpha}_z^k)^2. \tag{A10}$$

Position-averaging with the same slowly-varying assumption and $r_{p,x} = r\sin\theta\cos\phi$, $r_{p,y} = r\sin\theta\sin\phi$:

$$\langle \tilde{u}'_x \tilde{u}'_y \rangle = -\frac{N\sigma_x\sigma_y\sigma_x\sigma_y\sigma_z\,\tilde{\alpha}_z^2}{V_0}\int_0^1 r^4(1-r^2)^2\mathrm{d}r \times \int_0^\pi \sin^3\theta\,\mathrm{d}\theta \int_0^{2\pi} \sin\phi\cos\phi\,\mathrm{d}\phi. \tag{A11}$$

The azimuthal integral $\int_0^{2\pi}\sin\phi\cos\phi\,\mathrm{d}\phi = 0$, so $\langle \tilde{u}'_x \tilde{u}'_y \rangle = 0$. By cyclic symmetry the same applies to $\langle \tilde{u}'_y \tilde{u}'_z \rangle$ and $\langle \tilde{u}'_z \tilde{u}'_x \rangle$, so the principal-frame second moment is exactly diagonal and Equation (A9) together with its y, z-analogues fully fix R.

Inversion. Identical calculations for $\langle \tilde{u}'_y \tilde{u}'_y \rangle$ and $\langle \tilde{u}'_z \tilde{u}'_z \rangle$ give a 3×3 linear system in $\tilde{\alpha}_x^2, \tilde{\alpha}_y^2, \tilde{\alpha}_z^2$,

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} \tilde{\alpha}_x^2 \\ \tilde{\alpha}_y^2 \\ \tilde{\alpha}_z^2 \end{pmatrix} = \begin{pmatrix} \lambda_x/\tilde{C}_x \\ \lambda_y/\tilde{C}_y \\ \lambda_z/\tilde{C}_z \end{pmatrix}, \tag{A12}$$

with inverse

$$\frac{1}{2}\begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}. \tag{A13}$$

Substituting $1/\tilde{C}_i = \frac{315\,V_0}{8\,N\,V_e\,\sigma_i^2}$ gives, for $\beta \in \{x, y, z\}$,

$$\tilde{\alpha}_\beta^2 = \frac{315\,V_0}{16\,N\,V_e}\left[-\frac{\lambda_\beta}{\sigma_\beta^2} + \frac{\lambda_{\beta+1}}{\sigma_{\beta+1}^2} + \frac{\lambda_{\beta+2}}{\sigma_{\beta+2}^2}\right], \tag{A14}$$

indices taken cyclically.

Rebaptisation. The geometric prefactor in Equation (A14) factorises into a per-eddy piece and a global piece. Writing $\tilde{\alpha}_\beta^k = C_2\,c_{1,k}\,\alpha_\beta^k$ with

$$c_{1,k} = \frac{1}{\sqrt{V_e^k}}, \qquad C_2 = \sqrt{\frac{315}{16}\frac{V_0}{N}}, \tag{A15}$$

splits off the clean (tilde-free) intensity

$$\alpha_\beta = \left[-\frac{\lambda_\beta}{\sigma_\beta^2} + \frac{\lambda_{\beta+1}}{\sigma_{\beta+1}^2} + \frac{\lambda_{\beta+2}}{\sigma_{\beta+2}^2}\right]^{1/2}, \tag{A16}$$

which is Equation (7). Substituting back into Equation (A2), the global C_2 pulls out of the eddy sum and the per-eddy $c_{1,k}$ remains inside:

$$\tilde{u}'_\beta(\boldsymbol{x}) = C_2 \sum_{k=1}^{N} c_{1,k}\,\sigma_\beta^k\,q(d^k)\,(\boldsymbol{r}_p^k \times \boldsymbol{\alpha}^k)_\beta, \tag{A17}$$

which, on rotation by Equation (A1), reproduces Equations (3) and (4) exactly. The constants in Equations (9) and (10) are therefore uniquely fixed by the variance-matching constraint and the eddy-shape integrals; the user-tunable prefactor s in Equation (10) is a scale parameter that can be used to alter the turbulence intensity.

[1] Poletto, R., Craft, T., & Revell, A., *A new divergence-free synthetic eddy method for the reproduction of inlet flow conditions for LES*, Flow, Turbulence and Combustion, 91(3), 519–539, 2013. DOI: 10.1007/s10494-013-9488-2.

[2] Fyhn, H., Gruber, A., Nygård, H. T., Dawson, J., & Nogenmyr, K.-J., *Numerical and Experimental Investigation of a Non-Premixed Flame of Partially Decomposed Ammonia at Atmospheric Pressure*, Journal of Engineering for Gas Turbines and Power (submitted, 2026).

[3] Fyhn, H., Meyer, O. H. H., Wang, Q., Worth, N., Koomen, J., Dammers, T., & Gruber, A., *Numerical and Experimental Investigation of a Laboratory-Scale Hydrogen-Fired FlameSheet^{TM} Burner Operated at Atmospheric Pressure With Full Optical Access*, Journal of Engineering for Gas Turbines and Power (submitted, 2026).